

Linear Algebra Assignment

Name: Yash Raj

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Question #1:

$$x - 4y = 1$$

$$2x - y = 3$$

$$-x - 3y = 4$$

Do these lines have common point of intersection?

Sol:-

$$x - 4y = 1 \rightarrow (i)$$

$$2x - y = 3 \rightarrow (ii)$$

$$-x - 3y = 4 \rightarrow (iii)$$

Taking eq (i) & eq (ii)

$$x - 4y = 1$$

$$x = 1 + 4y \rightarrow (iv)$$

put in eq (ii)

$$2(1 + 4y) - y = 3$$

$$2 + 8y - y = 3$$

$$x - 4\left(-\frac{5}{7}\right) = 1$$

$$\boxed{y = -5/7}$$

$$\boxed{x = -13/7}$$

put in eq (iii) the values of x & y:

$$-(-13/7) - 3(-5/7) = 4$$

$$13/7 + 15/7 = 4$$

$$\left(-\frac{13}{7}, -\frac{5}{7}\right)$$

Ans

$$4 = 4$$

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find Equation involving g, h, k
that makes augmented matrix consistent
system.

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{bmatrix}$$

Soln

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & -3 & 5 & 2g+k \end{bmatrix}$$

if $(2g+k+h=0)$ then system will
be consistent because $\text{Rank}(A) =$
 $\text{Rank}(B)$

Ans

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Question #3:

$$w + x + y + z = 6$$

$$w + x + z = 4$$

$$w + y = 2$$

- (a) List the leading variable.
- (b) List free variable.
- (c) The general solution of system is:

(a) leading variables $\Rightarrow w, x, z$.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 1 & 0 & 1 & 1 & 4 \\ 1 & 0 & 1 & 0 & 2 \end{array} \right]$$

w and $x, z \Rightarrow$ leading variables y .

(b) free variable $\rightarrow y$.

$$\begin{aligned} (c) \quad w + x + y + z &= 6 \rightarrow (i) \\ w + y + z &= 4 \rightarrow (ii) \\ w + y &= 2 \rightarrow (iii) \end{aligned}$$

Subtract (3) from (2).

$$w + y - w + z - y - z = 2 - 4$$

$$z = 2 \text{ put in eq (i)}$$

$$w + x + y + 2 = 6$$

$$w + x + y = 4 \rightarrow (5)$$

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2) Subtract eq (1) from eq (2) :

$$w + n + y - w - y - z = 4 - 6$$

$$\boxed{n = 2}$$

$$n = 2, z = 2$$

$$w + y + z = 4$$

$$w + y = 2$$

$$\boxed{w = 2 - t}$$

$$y = t$$

free variable

$$(w, n, y, z) = (2 - t, 2, t, 2) \text{ for all } t \in \mathbb{R}$$

$$(d) -2w - 2y = -3$$

$$-2(w + y) = -3$$

$$\boxed{w + y = 3/2} \rightarrow \text{new equation}$$

$$w + y = 2 \rightarrow \text{old equation}$$

if we include new equation then it will contradict (false) previous equation

$\rightarrow$ so solution doesn't exist.

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Question #4

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 2z &= 11 \\2x + 3y - 4z &= 3\end{aligned}$$

$$\begin{aligned}x + y + z &= 7 \\x + 2y + 2z &= 10 \\2x + 3y - 4z &= 3\end{aligned}$$

(a) Solve using Gauss Jordan method:

Augmented matrix:

$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 1 & 2 & 2 & 11 & 10 \\ 2 & 3 & -4 & 3 & 3 \end{array} \right] \begin{array}{l} \\ R_2 - R_1 \\ R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 1 & -6 & -9 & -11 \end{array} \right] R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 0 & -7 & -14 & -14 \end{array} \right] \begin{array}{l} -1R_3 \\ 7 \end{array}$$

$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right] R_2 - R_3$$

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$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right] R_1 - R_3$$

$$\sim \left[\begin{array}{ccc|cc} 1 & 1 & 0 & 4 & 5 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right] R_1 - R_2$$

$$\sim \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 4 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right]$$

$[x=1, y=3, z=2] \rightarrow \text{System I}$

$[x=4, y=1, z=2] \rightarrow \text{System II}$

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(b) Solve the system using inversion algorithm.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 3 & -4 \end{bmatrix} \quad [X = A^{-1}b]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 3 & -4 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -6 & -2 & 0 & 1 \end{array} \right] R_2 - R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & -7 & -1 & -1 & 1 \end{array} \right] \frac{-1}{7}(R_3)$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \end{array} \right] R_2 - R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{8}{7} & \frac{6}{7} & \frac{1}{7} \\ 0 & 0 & 1 & \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \end{array} \right] R_1 - R_3$$

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$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 6/7 & -1/7 & 1/7 \\ 0 & 1 & 0 & -8/7 & 4/7 & 1/7 \\ 0 & 0 & 1 & 1/7 & 1/7 & -1/7 \end{array} \right] R_1 - R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -8/7 & 4/7 & 1/7 \\ 0 & 0 & 1 & 1/7 & 1/7 & -1/7 \end{array} \right]$$

$$B_1 = \begin{bmatrix} 6 \\ 11 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 4/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 3 \end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|c} 12 & -11 & 3 & 1 \\ -48/7 & 60/7 & 3/7 & 3 \\ 6/7 & 11/7 & 3/7 & 2 \end{array} \right]$$

$$(x, y, z) = (1, 3, 2)$$

$$\begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 4/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 14 & -10 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$(x, y, z) = (4, 1, 2)$$

B

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Question #5:

$$C = \begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 1 & 0 & a \end{bmatrix}$$

(a) find $\det(C)$ as function of a .

$$a(a^2 - 0) - 1(0 - 1) + 0 = a^3 + 1$$

$$\det(C) = a^3 + 1$$

$$(a+1)(a^2 - a + 1)$$

(b) for which value of a is invertible.

$$a^3 + 1 \neq 0$$

$$a^3 \neq -1$$

$$a \neq -1, a \neq \frac{1 + \sqrt{3}i}{2}, \frac{1 - \sqrt{3}i}{2}$$

$$Cx = b$$

- if $\det(C) \neq 0$ ($a \neq -1$), then C is invertible.
- if C is invertible then system has unique solutions.

$\Rightarrow$ if $\det(C) = 0$, ($a = -1$) then C is not invertible.

in this case system could have no solution or infinitely

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Question #6:

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x + 2y, y)$$

$$(a) T(e_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T(e_2) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

(b) Interpret geometrically:

A is upper triangular with 1's on diagonal & non-zero below it.

So then $\det(A) = 1$

The transformation leaves y -co-ordinate unchanged & it shifts x -co-ordinate & proportional to y .

Therefore T is (horizontal shear)

$$(c) (0,0) (1,0) (1,1) (0,1)$$

$$A^n = \begin{bmatrix} 1 & 2n \\ 0 & 1 \end{bmatrix}$$

$$A^n(0,0) = (0,0) \quad A^n(1,0) = (1,0)$$

$$A^n(1,1) = (1+2n, 1) \quad A^n(0,1) = (0,1)$$

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$$\begin{aligned} a_n + by - 3z &= -3 \\ -dn - by + 12z &= 1 \\ a_n + by - 12z &= -3 \end{aligned}$$

has solution $[x=1, y=1, z=2]$

eq (i) $a - b - 0s - 3 = 0 \Rightarrow a - b = 3 \rightarrow (i)$
eq (ii) $-2 + b + 2a = -1 \Rightarrow b + 2a = 1 \rightarrow (ii)$
eq (iii) $a - 3 - 2c = 0 \Rightarrow a - 2c = 3 \rightarrow (iii)$

$$a = 3 + b \rightarrow (4) \quad a = (1-b)/2 \rightarrow (5)$$

put in (3)

$$\begin{array}{l} 3+b=2c \\ 3+b \neq x(1-b) \quad \Rightarrow 3+b=1-b \\ x \qquad \qquad \qquad 2b=-2 \\ b+2c=1 \rightarrow (2) \quad [b=-1] \end{array}$$

2052
1051

9526
[952]

$$(a, b, c) = (2, -1, 1) \text{ Pm}$$

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Question #8.

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(i) Because pivot position tells us that which variable is leading in that row.

(ii) if a co-efficient method matrix has pivot in every row, it means each equation contributes to leading variable.

(iii) Inconsistency occurs only when row in reduced echelon form is $(0=d)$ it creates contradiction.

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Question #9,

$$\begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

(a) find inverse using inversion algorithm & verify if exist.

matrix

$$\sim \begin{bmatrix} 5 & 3 & 2 & : & 1 & 0 & 0 \\ 3 & 3 & 2 & : & 0 & 1 & 0 \\ 0 & 1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \xrightarrow{1R_1}$$

$$\sim \begin{bmatrix} 1 & 3/5 & 2/5 & : & 1/5 & 0 & 0 \\ 3 & 3 & 2 & : & 0 & 1 & 0 \\ 0 & 1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 3R_1}$$

$$\sim \begin{bmatrix} 1 & 3/5 & 2/5 & : & 1/5 & 0 & 0 \\ 0 & 6/5 & 4/5 & : & -3/5 & 1 & 0 \\ 0 & 1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \xrightarrow{5/6 R_2}$$

$$\sim \begin{bmatrix} 1 & 3/5 & 2/5 & : & 1/5 & 0 & 0 \\ 0 & 1 & 2/3 & : & -1/2 & 5/6 & 0 \\ 0 & 1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - 3R_2}$$

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$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] R_2 \leftarrow R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] R_3 \leftarrow R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 1 & 1 & 3/2 & -5/2 & 3 \end{array} \right] R_2 \rightarrow R_2 - \frac{2}{3}R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1 & 0 & -3/2 & 5/2 & -2 \\ 0 & 0 & 1 & 3/2 & -5/2 & 3 \end{array} \right]$$

$$\sim \begin{array}{l} 1 \\ 2 \\ 3 \end{array} \left[\begin{array}{ccc} 1 & -1 & 0 \\ -3 & 5 & -4 \\ 3 & -5 & 6 \end{array} \right]$$

(b) find inverse of matrix 2:

$$\left[\begin{array}{ccc} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{array} \right] \quad \begin{array}{l} \text{no inverse exist.} \\ \text{det: } 0. \end{array}$$

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(b)

$$\sim \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 + 4R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 0 & 0 \\ 0 & -10 & 7 & -4 & 0 & 0 \end{array} \right] R_3 + R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 & 0 \end{array} \right] \frac{1}{10} R_2$$

 $\det(A) = 0$ $\rightarrow$ no inverse exists.

$$\sim \left[\begin{array}{ccc} 1 & -3 & 4 \\ 0 & 1 & -7/10 \\ 0 & 0 & 0 \end{array} \right] R_1 + 3R_2$$

$$\sim \left[\begin{array}{ccc} 10 & 0 & 19/10 \\ 0 & 1 & -7/10 \\ 0 & 0 & 0 \end{array} \right]$$

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(b) Solution of system $Ax=0$.

(i) Matrix 1:-

for an invertible matrix,
the only solution $Ax=0$ is trivial
solution.

$$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(ii) Matrix 2:-

$$\text{RREF: } \begin{bmatrix} 1 & 0 & 11/10 \\ 0 & 1 & -7/10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow x + \frac{11z}{10} = 0$$

$$y - \frac{7z}{10} = 0$$

$z = t$ (free variable)

So equation becomes

$$x = -\frac{11t}{10}, y = \frac{7t}{10}, z = t$$

$$x = t \begin{bmatrix} -11/10 \\ 7/10 \\ 1 \end{bmatrix}$$

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(c) find solution Ans for $b = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$

Soln:-

$$Ax = B$$

$$x = A^{-1}B$$

(i) for matrix:-

$$A = \begin{bmatrix} 1/2 & -1/2 & 0 \\ -3/2 & 3/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -1 & 0 \\ -6 & 15 & -10 \\ 6 & -5 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 \\ 11 \\ 18 \end{bmatrix}$$

(ii) for matrix 2:-

→ for matrix 2, inverse doesn't exist & last row is contradictory ($0, 0, 0 = c$)
 $c \neq 0$.

→ No solution, so the system Ans b is inconsistent.

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Question #10

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$$u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$b = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3 \text{ by}$$

$$T(x) = (x_1 - 3x_2, 3x_1 + 5x_2, -x_1 + 7x_2)$$

$$(a) T(e_1) = T(1, 0) = (1, 3, -1)$$

$$T(e_2) = T(0, 1) = (-3, 5, 7)$$

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$$

$$(b) T(u) \text{ for } u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$T(x_1, x_2) = (x_1 - 3x_2, 3x_1 + 5x_2)$$

$$T(2, -1) = (5, 1, -9)$$

$$T(u) = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

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(c) Ans b for $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$

Soln,
Solving by substitution.

$$x_1 - 3x_2 = 3$$

$$3x_1 + 5x_2 = 2$$

$$-x_1 + 7x_2 = -5$$

$(x_1, x_2) = (3/2, -1/2)$ put both in eq (i)

$$\frac{-3}{2} + 7(-1/2) = -5$$

$$-5 = -5$$

$$x = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix}$$

$$x_1 = 3 + 3x_2$$

put in eq (ii)

$$3(3 + 3x_2) + 5x_2 = 2$$

$$9 + 9x_2 + 5x_2 = 2$$

$$x_2 = -1/2$$

$$x_1 = 3 + 3(-1/2)$$

$$x_1 = 3/2$$

(d) Is there more than one x where image under T is b ?

Ans:

No, since, the determinant of $A \neq 0$ & both equations are linearly independent. So solution is always unique.

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(c) Determine if c is Range of T .
like from eq (i) (ii) & (iii) we get.

$$x_1 = 3/2, \quad x_2 = -1/2$$

$$C = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$x_1 - 3x_2 = 3$$

$$3x_1 + 5x_2 = 2$$

$$-x_1 + 7x_2 = 5$$

→ Here we can see that the last equation will contradict the solution $5 \neq -5$

→ No solution exist.

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Question #12.

find standard matrix A for linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T(x) = Ax = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix}$$

$$A\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} a+b \\ c+d \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$a+b=1 \rightarrow (i)$$

$$c+d=-2 \rightarrow (ii)$$

$$A\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2a+3b \\ 2c+3d \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

$$2a+3b=-2 \rightarrow (iii)$$

$$2c+3d=5 \rightarrow (iv)$$

Solve for a & b, c & d:

$$a+b=1, \quad a=1-b \rightarrow (i)$$

$$c=-2-d \quad c=-2-d \rightarrow (ii)$$

put $c(v)$ in $c(iii)$.

$$2(1-b)+3b=-2$$

$$b=-4$$

$$2-2b+3b=-2$$

$$a=5$$

put value of c in $c(iv)$

$$2(-2-d)+3d=5$$

$$-4-2d+3d=5$$

$$[d=9] \text{ put in } (i) \rightarrow c=-11$$

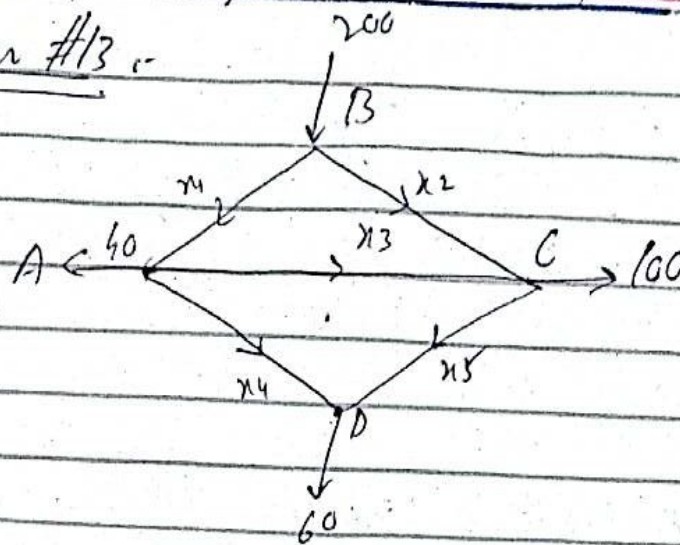
$$\begin{bmatrix} 5 & -4 \\ -11 & 9 \end{bmatrix}$$

Ans

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Question #13:-



Nodes	flow-in	flow out
A	x_1	$x_4 + x_3 + 40$
B	200	$x_1 + x_2$
C	$x_2 + x_3$	$x_5 + 100$
D	$x_4 + x_5$	60

Rearranging:-

$$x_1 + x_2 = 200 \rightarrow (i)$$

$$x_1 = 40 + x_3 + x_4 \rightarrow (ii)$$

$$x_2 + x_3 = 100 + x_5 \rightarrow (iii)$$

$$x_4 + x_5 = 60 \rightarrow (iv)$$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 200 \\ 1 & 0 & -1 & -1 & 0 & 40 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right] R_2 - R_1$$

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 200 \\ 0 & -1 & -1 & -1 & 0 & -160 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right] -1R_2$$

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$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 160 \\ 0 & 1 & 1 & 0 & 100 \\ 0 & 0 & 0 & 1 & 60 \end{array} \right] \quad R_3 - R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 160 \\ 0 & 0 & 0 & -1 & -60 \\ 0 & 0 & 0 & 1 & 60 \end{array} \right] \quad -1(R_3)$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 160 \\ 0 & 0 & 0 & 1 & 60 \\ 0 & 0 & 0 & 1 & 60 \end{array} \right] \quad R_4 - R_3$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 160 \\ 0 & 0 & 0 & 1 & 60 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + x_2 = 200 \rightarrow (1)$$

$$x_2 + x_3 + x_4 = 160 \rightarrow (2)$$

$$\text{Let } x_3 = s \text{ \& } x_4 = t$$

$$x_1 = 40 + t + s$$

$$x_2 = 160 - s - t$$

$$x = s$$

$$x_4 = t$$

$$x_3 = 60 - t$$

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(b) if x_4 is closed.

$$x_4 = 0, t = 0.$$

$$x_1 = 40 + s, x_2 = 60 - s, x_3 = s, x_4 = 0, x_5 = 60$$

(c) Minimum value of x_1

$$x_1 = 40 + s$$

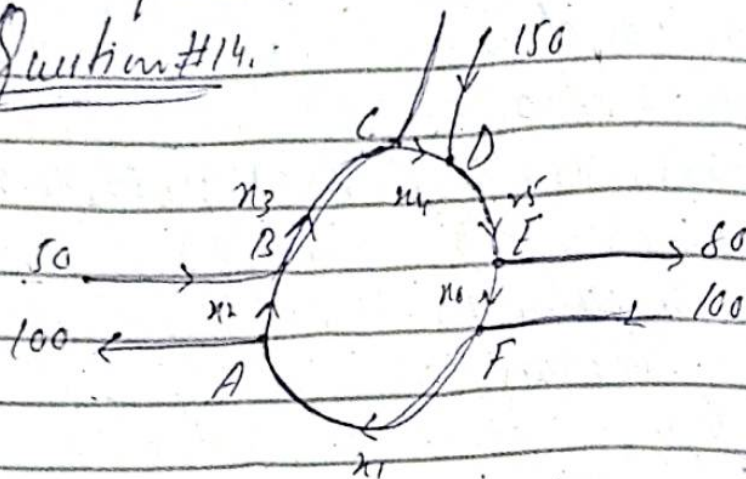
So

$$\boxed{x_1 = 40} \text{ minimum } x_1$$

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Question #14.



Node	flow in	flow out
A	x_1	$x_2 + 100$
B	$x_2 + 50$	x_3
C	x_3	$x_4 + 120$
D	$x_4 + 150$	x_5
E	x_5	$x_6 + 80$
F	$x_6 + 100$	x_1

$$x_1 - x_2 = 100 \rightarrow (i)$$

$$x_2 - x_3 = -50 \rightarrow (ii)$$

$$x_3 - x_4 = 120 \rightarrow (iii)$$

$$x_4 - x_5 = -150 \rightarrow (iv)$$

$$x_5 - x_6 = 80 \rightarrow (v)$$

$$x_6 - x_1 = -100 \rightarrow (vi)$$

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$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & 0 & 0 & 0 & 1 & -100 \end{array} \right] \quad R_6 \leftarrow R_6 + R_1$$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \quad (R_2 + R_1) \text{ on } R_6$$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & -1 & 0 & 0 & 1 & -50 \end{array} \right]$$

again same step:-

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} x_1 - x_2 = 100 \\ x_2 - x_3 = -50 \\ x_3 - x_4 = 120 \\ x_4 - x_5 = -150 \\ x_5 - x_6 = 80 \end{array}$$

$\Rightarrow$ x_6 is a free variable because last row is 0.

Question #18.

$$\Delta_s \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}, (b-a)(c-a)(c-b)$$

(i) Subtract first column from second & third.

$$\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$$

(ii) Expand along first row.

$$\Delta_s 1 \cdot \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix}$$

$$\Delta_s \begin{vmatrix} b-a & c-a \\ (b-a)(b+a) & (c-a)(c+a) \end{vmatrix}$$

$$\Delta_s (b-c)(c-a) \cdot \begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix}$$

2) simply determinant of 2×2 :-

$$\Delta_s (b-c)(c-a) \cdot (c-b)$$

P3